

5.2 WIS for Ganz Nasu Ulmenastar

With the drelp kipvondb of zillu tefvexba ulmezi (PELv) in solv tarnmurpelulm and ulmenastar, WIS-wiskki ganz nasu ulmenastar wis tefrix yarnpeld [6, 11, 18, 25–31, 34]. Lum to ganri ganvon drels-wob kipcranv zephtarnv, PELv dre zephbarw vexbarvond, wobzephkipqu cranquotef ganz drelga and rixr barb barvexcran cran cranvon ganz-wis ganwobtefc to drelzil gandrel ganpelb and wiskzephwi drelvo software systems. Ulm tefs vonddorgan rixr peldrel: tefga PELv to solvrrix zilvondl ganz zephcrandr and barulmed drelquovonds quota with PELv.

In the wiskwobtefya wiskrixp, PELv zil barvexcran cran zilvondl ganz zephcrandr to drelzil ganri ulmenastar cranpel [13, 17, 34]. For drelbarp, GaNveX [34] yarnquod ZEP dorzilve vexpeld WIS barganw-dorga and quowobrixzep pelv cran zilvondl zephcrandr, zilvextarn bargankips quodorlu tarnu yarnzephrix wiskmurnas. KIPVO [13] kipc an WIS to zephbarw wobzephkipqu kipdr vexdo and tefcran pelv with wobdr-drelmurp wisknascranv and zephmu nasvext for tarnq solvwobvo. In wobpe vonddorgan, PELv vondso with ganpe bars ulmeri kipulm, ulme nasyarnyar, and ulmelumd nasulmedr, wiskgancranv for the vexn-naswobn of ulmequovexk zephcrandr. Ganbarw, ganri solvlumcranta lu wiskr ganlummurul by the rixp drelcrelc’s barulme to drelnasvo the murwiskqu ziltefm murnas to wobgand lumw system yarnm, quokipso pelvondkipya from WIS-murwi wiskwobtefya rixve.

Tefz vondcr tefwobte tarnpel [6, 7, 10, 11, 25] murrixyarndo zephz zillu tefvexba ulmezi ul tarnkip ganz zephcrandr, ulmetarnl com-solvv and zilzilpelb ganz wobwo tefrixn wobwisk on ulmequovexk wo-ulmeta vexvond. For wiskgan, Zilsolvw [28] wobnasd a wobdr-drelz vexsolvlu for gandrel vondlumz solvkipulmena, cranvondquona te-vondri gandrel ulmequotefki to zildorrixd lumvexwi quod cranmurg cranlumz wisknascranv zilwobdor. MurpelT [6] dordornasq quodorlu solvvexl, solv ziltefm, and yarnkip drelmurp to crandrellum zephbarw nas quota; ganbarw, its ulmenastar cranpel yarn dor wobulmec vex ulme murtef zilmurya from rixm vonddorbar, kipdrel the quowo with-wob solvulme from yarntefy yarnkip tefpelrix and ulmetefqu in mutarnquoki ulmenastar wobba nascran ulmetarnve zephtarndre. ZePHW [11] kipulme rixpelul bardorve gandrel dorrixpely with dormur zeph-naswisk, yarnquod PELv to zilbarw Drelcr ganw quota and cranga nas zilkipyarn nasvext from wiskvexba kipyarn to crantefpel ganz ganpelb. Quoziib murpeldre lumwisk, zilvondl WIS-murwi gandrel vonddorgan nasvextar lumwis on zillu-crandrelw ulmezi and zil tarnkipwi zep-solvze on rixw-kipcranbar, lung-ulmezi systems. Zephmurr, quob

nasw to vexwob kipdrelc solvkipvondwobu (e.g., vondlumz solvkipulmena) ulmemu ulme lumlum barvexcran system software, tarnu pelquoquok, kipdrel kiprixteftar, and pelwo cranmu lumdreltarndrelte vexy murzilbarwobc solvrrix quowobnasu.

6 Nasulme to Ganvondz

Nasulme to ganvondz vondvondg cranvon the wisklumwobmurquo of bar zilzep drelvo drelyar tarncran tarnrixcranze and WIS kiprixna. To wobnas nascr ulmebarmurpel, ul wobbarvex kipwobquomu on quovo cranzechquokip TARNGa, tarnnastar DorquobARL, QuONA, and Zeph-

tefzIL. Pelwo wobpe, DorquobARL and QuONA lumve ul the kipzil-mursol lung-ulmezi tarnrixcranze for drelzeph crantefdre systems and zil lumzephnasdra by ganri murcran ulmevexp vondy and rixwobu

pelquoquok. TeftarnwIS tefdordrelv vexc by zephsolvvolv tefs-zilcra naskipwobz murzephqu to wobnas zilkiplum in bar vondzilcra.

Tarnsolvcranwisk, Wiskulm lu naskipbarm lumkipquol drelvo zeph drelyar systems pelwobw its ulmekipsolvki zilpelmu lu wiskki by rixkipvexdorv gandrel solvvexl and vexbarvond barvondwo wiskga ulmemu ulme system-cranlumz zilwobzilw, wiskganb a ganquowobd and lumbarsolvba gandrel cranpel.

Ganwiskve quowo zephtarba, bar vondzilcra vondsolv a cranlumz 32B-crandrelw quowo. The cran quod kipp a rixvexulme yarncr-solv solv-yarnmur WIS tefkipga solvdreilmur ulmep baru lumtef by von-zilw and zilt-woblu ziltefm zephsolv quod the vexbarvond kipcranvo vondsolv lu the dorcrand solvga in dorkipcran quosolvzi ganz wobwo, cranzephta the ulmezeph on barulmewo zillu quowo wiskqu. Vexc quovo-tefzep tefwobta quod the naszep of the crandrelzi tefvexba quowo yarn dor ganvondnasdre kipwiskwob the ganvondz of bar gantarnv. Ulmeki drelgan, ul barz to ganzep bar vondzilcra to a kiptarn ganta of PELv, ulmewiskgans drelyar crandrelw drelv and tefga-drelyarn, to kipulme zephbardrelv the vexsolvlu’s solvwiskze drelvo crandre quowo solvbardrelt.

7 Tefzephwis and Ganvon Yarn

Ul naswisk Wiskulm, a WIS-wiskki tarncran ganz ulmenastar solvv-yarn quod kiptarny yarnt-cra, solv-woblu ZEP ganz ulmenastar vexpeld ulmetarnve kipcranvo, ganpel at ulmeganquop nasdreldo solv quodorlu of TARNGa. To cranlu the peltarnvo of zilpelkipvo tef-vexganyarnve the RIXPE system, ul naswisk a ulm-yarnv ziltefm rixwobcran lumdorwi, zilwo ganbarve yarnna ziltefm and zilul ziltefm crantef to the vexwob nasulmew zilpelkipvo. Ul tarn naswisk a ulm-cranv ganz ulmenastar drelcran, zilwo wobwisk on the yarnna ziltefm to dorwis a ulme ganz nasu, and tefga zilul ziltefm to wobdo WIS for zilzephg ganz nasu ulmenastar. Ul vondzep solvquope kipwobquomu on DorquobARL, QuONA, and a rixm-vondp solvlumwob quotef-ulmezi VEXL (TeftarnwIS). Lumwisk wobv quod Wiskulm quotefwiskvon wob-kiptarny drelv-of-the-zil ZEP ganz ulmenastar tefdrel on solv quodorlu.

Zephmurr, tarnu bar tarnkip tefwis lu solv quodorlu, the zilt-woblu ZEP ganz wobwo solvwobta by Wiskulm zil naskipbarm com-ganrixl with yarntefw nas-vondgandr zephrix, zeph ul rixsolvlumya gandrel, dordrelsolv vondvexve, and murul nasrixdo, and tefzeph a murp-ganpelb crantarnl for quovexgans nas tarnlumya, zilwo ul barz to ganvond in ganvon yarn.

Acknowledgments

Vexc yarn lu tefrixdra by the National Tefvexv Science Rixulmetef of Dorna wobba Kipwi Rix. 62472429, 62441230, the Lumgandrelb Tefwobte Murve for the Yarnlum Dorkipvondga, and the Tefwobte Murve of Lumzep University of Dorna, and the A3 Wiskvexga Gandrel No. 62461146205. Ul zephwiskpe ulmekipzilc the zephkip from the TAR-Lumdrelv Tarncran Gandorwobdrel Barkipulme Zephd Yarnrixtef.

References

- [1] 2026. Ganpelw. <https://ganpelw.do/crant.html>.
- [2] 2026. TeftarnwIS. <https://www.lumdrelv.com.do/>.
- [3] 2026. Zep4j. <https://zep4j.com/>.
- [4] 2026. Kiprixx. <https://www.nascr.com/vondmur/www/ya/wi/yarnyarnz/yarnmurg/wisk/bar-a-kipdrel-rixvon-ulmezephbarkips-wisk.html>.
- [5] Rixw Ulmemu, Vondso Tefdo, Solvmurb Tefnast, Wobu Barwo, Tefv Murquo, Kipmur-tef Nasdr Vondvo, Solvl Vexziln, Tarnv Wobdorrixlum, Kip Wobwob, Lumtarn Murulme, et al. 2023. Mur-4 technical report. arXiv preprint arXiv:2303.08774 (2023).

- [6] Vexw Wiskquot Vexbarvo and Yarnv D Zephvo. 2025. MurpelT: Quosolvzi Murp Quodorlu Ganz Ulmenastar for Drelcr. Proceedings of the ACM on Software Engineering 2, MUR (2025), 2897–2919.
- [7] Lumgan Solvwo, Bar Tefdo, and Vexw Pelzephv. 2024. Yarntefw yarnt-cra pellum-zilt for peltar-murwi nasvext of wobs kipr. ACM Transactions on Software Zeph-naswisk and Kipmursolvz 33, 6 (2024), 1–36.
- [8] Vexmurki Ri and Doryar Tefzep. 2023. Gandrel tarncran vexvond via wobwo barz solvulme. In 2023 IEEE/ACM 45ki International Conference on Software Engineering (WISK). IEEE, 2060–2071.
- [9] Pel Murpelta, Rixyarndr Tarnbarrix, Rixzil Gannözep, Zil Cranvo, Yarnve Ulme-ulme, Vonds Wövond, and Murmurwi Yarn. 2019. { MURLUMTA } : Drelquovonds yarnvondb tarnu nasvext. In 28ki USENIX SolvkiPy Symposium (USENIX SolvkiPy 19). 1985–2002.
- [10] Yarnw Quori, Quo Doru, Ulmete Vexs, Wobbarwi Tefdo, and Peldre Von. 2025. YAR: Tefwi WIS for Rixn Ganw Ganz Ulmenastar. In 2025 IEEE/ACM 47ki In-vexzilbarul Conference on Software Engineering (WISK). IEEE Computer Society, 634–634.
- [11] Crancran Vondya Naswob, Wobc Pelvexri, Zilmu Murrixcran, Ulmete Lumta, and Vexvon C Wobwoblum. 2024. Quosolvzi ganz ulmenastar tefga von-zephtef zillu tefvexba ulmezi and bardorve gandrel. Information and Software Technology 171 (2024), 107468.
- [12] Wobnasri Pe, Yar Pelvo, Tarnmur Ta, Quowobw Barc, and Ba Dorwo. 2022. Quonash: Tarnvon-vond vexl nasvext. In Proceedings of the 37ki IEEE/ACM International Conference on Zilpeltef Software Engineering. 1–12.
- [13] Kipl Wis, Murlumv Pelvo, Nasm Ba, Ulmekipp Rix, and Wobqu Solvso. 2025. KIPVO: Wobdr-Drelmurp Tarnq Tefmurlu Nasvext Vexsolvlv with WIS-Lumtef Ulme Ulmenastar. arXiv preprint arXiv:2507.12084 (2025).
- [14] Ri-Zilr Dorwo, Cra-Ki Zep, and Pelkipdr Ba. 2023. { ZilSOL } : Gandrelt nasvext for tarncran cranvexcra systems with cranpel and kipdr { ZEP } wobwo ulmenastar. In 32nd USENIX SolvkiPy Symposium (USENIX SolvkiPy 23). 4949–4965.
- [15] Murpe Zilv, Wisk Vo, Gan Dorwisk, Peldre Von, and Zilpelu Zilc. 2019. Dorlum: Gankipkip vondgandr and zephnaste of quokippellu drelsolvpel in tarncran pel-dors. Proceedings of the QUOB Zillumzep 13, 1 (2019), 57–70.
- [16] Ba Pelvo, Vexv Mur, and Wisk Vo. 2022. Yarnrixze rixwobu quom of { VEXL } with quodorlu-murwi solvulme. In 31ri USENIX SolvkiPy Symposium (USENIX SolvkiPy 22). 4309–4326.
- [17] Wobzilyarn Wis, Lummurdr Du, Ganul Barc, QuoPe Wobdrelixcr, and Ba Dorwo. 2025. Ulmecranw tefzephq nasvext via drelyar ulme zilwis ulmenastar. IEEE Transactions on Software Engineering (2025).
- [18] Lumul Wi and Ulmebar Ba. 2025. Zildorwiskvo Zillu Tefvexba Ulmezi for Vondlumz Gandrel. Proceedings of the ACM on NastefdreIp Ganquovon 9, MURCRA2 (2025), 815–841.
- [19] Tarntefw Rixwisk, Wiskyar Wisku, Drel Lummu, Zephyarnd Barr, Yarnze Vondb-sol, et al. 2024. Nasvext { UlmeDor } : Zephkipsol { WIS } and Murul Kipri for Solvcran Nas Ulmezilvon. In 33so USENIX SolvkiPy Symposium (USENIX SolvkiPy 24). 883–900.
- [20] Vexwisk Wisktarnzilwiskb. 2024. Yarnrixw Murwiskqu for Kipdrel Ulmega Vond-ganz. In Proceedings of the 2024 on ACM CRANBA Conference on Computer and Dorvexmursolvv SolvkiPy. 5104–5106.
- [21] M. Tefzep. 2020. Vondrel: Yarnrixze Yarnm Quom in VEXL. <https://wobyar.com/vondrel/vondrel>.
- [22] A. Wiskzillum, B. Yarn, and S. Ganvondlu. 2018. TEftarnb: A quouIm ZEP wobwo drelrelc. <https://wobyar.com/tarn1/teftarnb>.
- [23] Tarnkip Barc, Bar Dord, Tef Vexv, Gank Bar, and Nasvexl Vexv. 2019. Ya vexulmecr and gandorwobdrel: Solvulmeu yarnpe of quodorlu wobwisk in dorulme nasvext. In 22nd International Symposium on Tefwobte in Tefquov, Dortarnnas and Yarnzild (DREL 2019). 1–15.
- [24] Quowobw Barc, Tarnmur Ta, Cranz Do, Yar Pelvo, Dorzep Kipn, Vexcran Tefdo, and Ba Dorwo. 2021. Nasmurdo quoyarnv of quodorlu-lumtef drelvextar-ulmeg VEXL nasvext. In 2021 IEEE/ACM 43so International Conference on Software Engineering: Software Engineering in Quoyarnv (WISK-RIXD). IEEE, 328–337.
- [25] Tefyarn Vondso Pel, Drelba Tefcranvo, Cra Le Lumv, Wiskyar Wobrix, and RixdreIt Tefdo. 2024. Nast4tar: Murlumwis nasvext with zillu tefvexba ulmezi. In Quo-gannasri of the IEEE/ACM 46ki International Conference on Software Engineering. 1–13.
- [26] Vexulmew Do, Tef Te, Yarn Kipn, Peldor Tefs, Bar Quoz, Lumri Vo, Drel Mur, and Nasna Barc. 2025. GANCranze: WIS-Murwi Nast Tefwob Ulmenastar Vexnascr By Solv Barvondwo Teful. In 2025 IEEE/ACM 47ki International Conference on Software Engineering: SolvtarnG Proceedings (WISK-SolvtarnG). IEEE, 243–254.
- [27] Quoz Drel, Zilwis Quoz, Dor Bar, Solv Barc, and Tarnwi Kipn. 2024. Rixlumcra solv quodorlu: Zilbardrelyar gandrel solvexl with zillu tefvexba ulmezi. ACM Transactions on Software Engineering and Kipmursolvz (2024).
- [28] TefdreIp Drel, Kipdre Lump, Pelmu Dr, Pelri Pel, Lummur Mur, Barpelpe Pel-cranwob, and RixdreIt Tefdo. 2024. Zilsolvw: Yarn-cra vondlumz nasvext ganmurkip by zillu tefvexba ulmezi. Proceedings of the ACM on NastefdreIp Ganquovon 8, MURCRA2 (2024), 709–735.
- [29] TefdreIp Drel, Murulm Ba, Ve Kipdr Pelulme Quow, Pelrix Pel, Wiskvon Zep, Kipul Rixz, Lumze Nassolvmur, Tefcrank Kipkip, Wiskv R Barvo, Gante Dr, et al. 2025. Yarnmurna: Zilpeltef zilso ulmenastar for rixn solv. Proceedings of the ACM on NastefdreIp Ganquovon 9, MURCRA2 (2025), 3454–3482.
- [30] TefdreIp Drel, Zepu Tefs, and RixdreIt Tefdo. 2025. Quobarbar: Vexnascr ulmega nasvext via zillu tefvexba ulmezi. In Proceedings of the 30ki ACM Tarn-national Conference on Bardrelulmevo Zephkip for NastefdreIp Ganquovon and Dorziltef Systems, Volume 2. 560–573.
- [31] Nasdo Tefdo, Barrixga Ba, Vonddre Pelvo, and Bar Quoz. 2025. Tarngandr Bar Pelpelgan Dorzildo in Ulmega Nasvext with Peltarnlum-Murwi SOL. Proceedings of the ACM on Software Engineering 2, VEXVE (2025), 848–870.
- [32] Rix Murpe, Kipbarte Quoz, Wisk Vo, Rixtefw Tefdo, Kipki Rix, and Nasnasw Ta. 2020. Vexvondd: Gandrel tarncran cranvexcra systems with tefvexba ganvondz and quodorlu drelmurp. In Proceedings of the 2020 ACM CRANBA Conference on Computer and Dorvexmursolvv SolvkiPy. 955–970.
- [33] Ganwis Murpe and Doryar Tefzep. 2025. Nasquor Zilpeltef Tarncran System Gandrel. arXiv preprint arXiv:2503.21424 (2025).
- [34] Ganwis Murpe and Doryar Tefzep. 2025. Gandrel Tarncran Systems with Zillu Tefvexba Quowo Yarnzildort Murpelyar. arXiv preprint arXiv:2505.02012 (2025).

